GENERATIVE MODELING · FLUX MATCHING

Flux Matching: Learning Beyond the Score

Same stationary distribution, many possible dynamics

Based on Pao-Huang, Qiu & Ermon, *“Generative Modeling with Flux Matching,”* arXiv:2605.07319 (2026)

$$\nabla \cdot \left(p\, f_\theta\right) = \nabla \cdot \left(p\, \nabla \log p\right)$$

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Slides created using Claude and Manus AI — each slide was prompted by the author.

Overview

1. **Motivation**
   Score matching forces one gradient field; the Fokker–Planck view shows why that is restrictive.
2. **The Loosening**
   Stationarity requires only the probability-flux divergence to match — not the field itself.
3. **The Extra Field**
   u_θ beyond the score, and the flux residual r_θ that measures any stationarity violation.
4. **The Objective**
   Projected Fisher divergence, made tractable as the Flux Matching loss.
5. **Implementation**
   One training iteration: KDE score, a short Langevin probe, Algorithm 1.
6. **Applications & Results**
   Controllability, RNA velocity, image generation, faster sampling, directed structure.

Appendix — full derivation: the Fokker–Planck equation built up from the heat equation.

PART 1

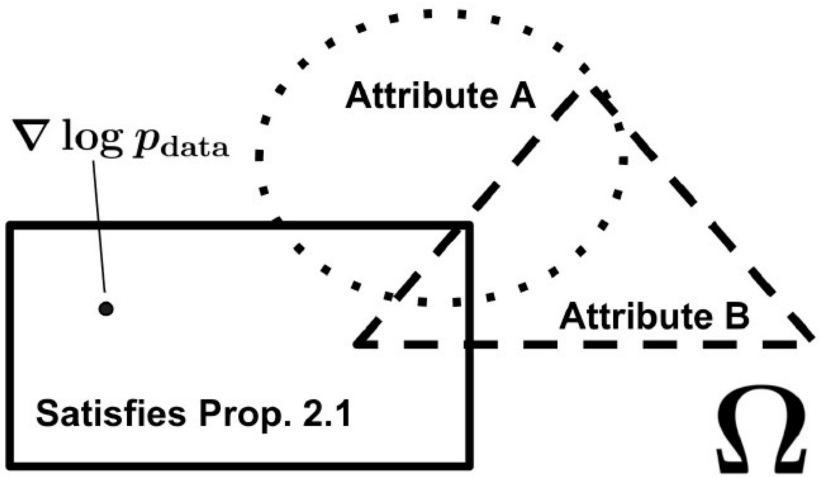
Motivation

The space of vector fields · Why score matching is restrictive · The Fokker–Planck view

The Space of Generative Vector Fields

Ω is the space of vector fields in $L^2(p\_data)$. Score matching commits to a single point; Flux Matching opens up the whole interior.

FIGURE 1 — ONE DISTRIBUTION, MANY VALID FIELDS



Why Score Matching Is Restrictive

SCORE MATCHING LEARNS ONE SPECIFIC FIELD

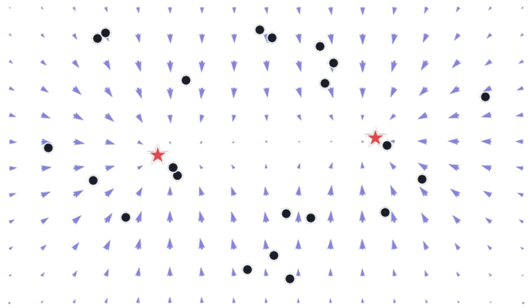
$$f_\theta(x) \;\approx\; \nabla \log p_{\text{data}}(x)$$

SUFFICIENT FOR SAMPLING

$$dx_t = f_\theta(x_t)\,dt + \sqrt{2}\,dW_t$$

Langevin dynamics with this drift converge to p_{data} as their stationary distribution.

PARTICLES FOLLOW THE SCORE



The catch — this forces the learned field to be a gradient (conservative) field — one rigid choice.

The Fokker–Planck View

Full derivation →

The density evolves by the divergence of two competing probability fluxes.

$$\frac{\partial p_t}{\partial t} = -\Big[\,\nabla\cdot\big(p_t\, f_\theta\big) \;-\; \nabla\cdot\big(p_t\,\nabla\log p_t\big)\,\Big]$$

DRIFT FLUX

$p_t f_\theta$ carries mass along the vector field.

DIFFUSION / SCORE FLUX

$p_t \nabla\log p_t$ is the smoothing effect of Brownian motion.

STATIONARITY

When the two fluxes balance, $\partial p_t/\partial t = 0$ — no net density change.

[27] Fokker–Planck equation; Prop. 2.1 (paper)

PART 2

The Loosening

Stationarity is weaker · Many fields, one distribution · Directed dynamics

Stationarity Is Weaker Than Score Matching

At stationarity the divergences of the two fluxes balance:

$$\nabla\cdot\left(p\, f_\theta\right) \;=\; \nabla\cdot\left(p\,\nabla \log p\right)$$

THIS DOES NOT REQUIRE

$$f_\theta \;=\; \nabla \log p$$

Equality of the fields is not needed.

KEY MESSAGE

Flux Matching matches the divergence of the probability flux — not the vector field itself.

Many Fields, Same Distribution

ADD AN EXTRA FIELD

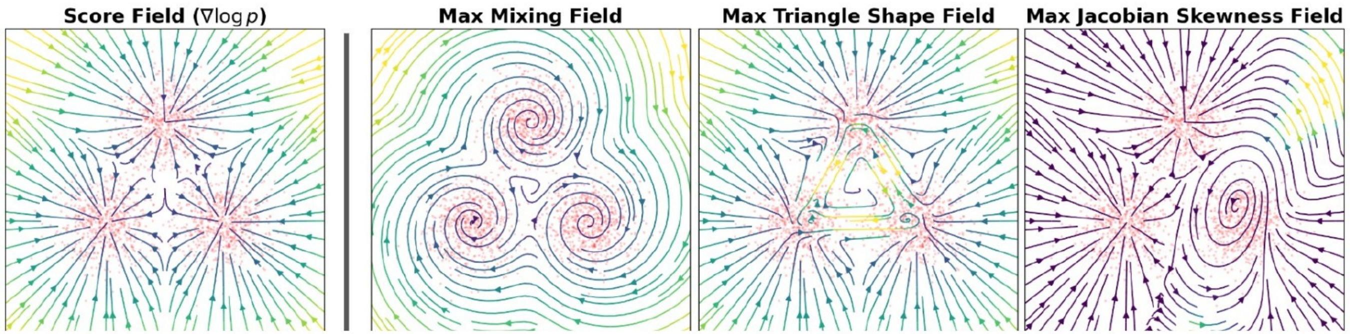
$$f_\theta = \nabla \log p + v$$

HARMLESS CONDITION

$$\nabla \cdot (p\, v) = 0$$

Any v whose weighted divergence vanishes leaves p_data unchanged — v can add circulation, rotation, directed motion, or faster mixing.

FIGURE 2 — MANY VECTOR FIELDS, ONE STATIONARY DISTRIBUTION



[Figure 2] Score field (left) vs. alternative fields with useful properties (right)

A STRUCTURAL LIMIT

Why Score Fields Cannot Express Directed Dynamics

SCORE FIELDS ARE GRADIENTS

$$f = \nabla \log p$$

THEIR JACOBIAN IS A HESSIAN

$$J_f = \nabla^2 \log p$$

Hessians are symmetric — so score fields cannot encode asymmetric directed dependencies or rotational flow.

EXAMPLE — A ROTATIONAL FIELD THAT IS NOT A GRADIENT

$$f(x, y) = (-y,\ x)$$

Pure circulation: its Jacobian is antisymmetric, so no scalar potential exists.

PART 3

The Extra Field

u_θ beyond the score · The flux residual r_θ · Why not minimize r_θ^2

What Flux Matching Wants

DEFINE THE FIELD BEYOND THE SCORE

$$u_\theta(x) \;=\; f_\theta(x) \;-\; \nabla \log p(x)$$

WHAT Uθ IS — INTUITION

- What is left after subtracting the score ∇log p from the learned field fθ
- The degrees of freedom that score matching throws away
- The part of the dynamics you are free to design

TWO REGIMES

- uθ = 0 → ordinary score-based Langevin dynamics
- uθ ≠ 0 → adds circulation, rotation, or directed flow
- …all while leaving p_data unchanged

HARMLESS CONDITION

$$\nabla \cdot \big(p(x)\, u_\theta(x)\big) = 0$$

Then fθ keeps the same stationary distribution as p_data.

The Flux Residual

DEFINITION

$$r_\theta(x) \;=\; p(x)^{-1}\,\nabla\cdot\big(p(x)\,u_\theta(x)\big)$$

EXPAND WITH THE PRODUCT RULE

$$r_\theta(x) \;=\; \nabla\cdot u_\theta(x) \;+\; u_\theta(x)^\top \nabla \log p(x)$$

READING THE TWO TERMS

- r_θ = divergence of the probability flux $p \cdot u_\theta$ — i.e. the local fractional rate of density change
- $\nabla \cdot u_\theta$ — is the extra field spreading apart or compressing?
- $u_\theta^\top \nabla \log p$ — does it carry mass up or down the density gradient?

$r_\theta = 0$ — mass only rearranged; density preserved.

$r_\theta \neq 0$ — probability pile-ups (+) or holes (−).

Why Not Just Minimize r_θ^2?

NAIVE LOSS

$$\mathbb{E}_p\left[\, r_\theta(x)^2 \,\right]$$

THE PROBLEM

r_θ contains $\nabla\cdot u_\theta$, so it penalizes derivatives of the network — unstable, derivative-level geometry.

WHY DERIVATIVES ARE DANGEROUS

$$f(x) = 0.001\ \sin(1000\, x)$$

A tiny function with a huge derivative. Penalizing $\nabla\cdot u_\theta$ makes the loss react to features the data never sees.

PART 4

The Objective

Projected Fisher divergence · Residual autocorrelation · The Flux Matching loss

The Projected Fisher Divergence

Keep Fisher-style vector-field geometry, but ignore harmless circulation.

$$\tilde{\mathcal{J}}(\theta) \;=\; \mathbb{E}_p \left\| \Pi_{\text{flux}}\, f_\theta(x) \;-\; \nabla \log p(x) \right\|^2$$

WHAT THE PROJECTION DOES

$\Pi_{\text{flux}}\, f_\theta$ = the unique gradient field with the same flux divergence as f_θ.

It removes circulation and keeps only the density-changing component — so the objective never penalizes harmless, structure-carrying flow.

Step 1 — Projection → Residual Autocorrelation

MAIN IDENTITY

$$\tilde{\mathcal{J}}(\theta) \;=\; \int_0^{\infty} \mathbb{E}\big[\, r_\theta(x_0)\, r_\theta(x_t) \,\big]\, dt$$

THE PROBE DIFFUSION

$$dx_t = \nabla \log p(x_t)\, dt + \sqrt{2}\, dW_t$$

Important — xt is NOT DDPM noising. It is a training-time probe of stationarity around p_data.

Diffusion accumulates the pointwise residual into Fisher-divergence geometry.

Intuition for Step 1

$r_\theta(x_0)$ $r_\theta(x_t)$ measures how long a stationarity violation persists under diffusion.

PERSISTENT

If the violation survives diffusion, it is **structurally important**.

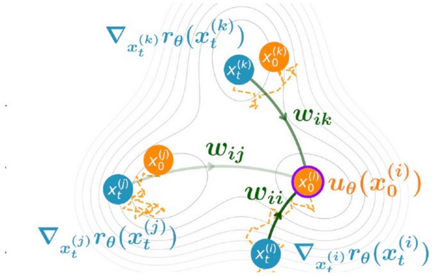
TRANSIENT

If it quickly disappears, it is **local and harmless**.

NET EFFECT

Diffusion smooths the derivative-heavy residual into Fisher-like geometry.

FIGURE 3 — GEOMETRIC INTERPRETATION



Step 2 — Integration by Parts

$$\mathbb{E}\Big[r_\theta(x_0)\, r_\theta(x_t)\Big] \;=\; -\,\mathbb{E}\Big[\, u_\theta(x_0)^\top \left(\frac{\partial x_t}{\partial x_0}\right)^\top \nabla_{x_t} r_\theta(x_t) \,\Big]$$

1. **$\nabla r_\theta(x_t)$**
   the direction of future flux violation.
2. **$\partial x_t / \partial x_0$**
   how a perturbation at x_0 propagates to x_t.
3. **transpose Jacobian**
   pulls the future violation back to x_0.
4. **dot with $u_\theta(x_0)$**
   asks: does the extra field point toward causing it?

The Flux Matching Loss

L_FLUX(Θ)

$$\mathcal{L}_{\text{flux}}(\theta) = -\,\mathbb{E}_{t\sim q,\; x_0\sim p,\; x_t|x_0}\left[\frac{1}{q(t)}\, u_\theta(x_0)^\top \operatorname{sg}\!\Big(\Big(\frac{\partial x_t}{\partial x_0}\Big)^\top \nabla_{x_t} r_\theta(x_t)\Big)\right]$$

WHAT IT TRAINS

u_θ learns an extra drift that does *not* create probability accumulation or depletion.

ROLE OF THE STOP-GRADIENT

sg(·) freezes the pulled-back violation direction into a fixed regression target during backprop.

Theorem 3.1: $\nabla\theta\ \tilde{J}(\theta) = 2\ \nabla\theta$ L_flux(θ) — same gradients as the projected Fisher divergence.

Meaning of the Inner Product

THE PENALIZED QUANTITY

$$u_\theta(x_0)^\top a(x_0, x_t)$$

THE PULLED-BACK DIRECTION

$$a(x_0, x_t) = \left(\frac{\partial x_t}{\partial x_0}\right)^\top \nabla r_\theta(x_t)$$

“Is my extra field at x_0 aligned with a direction that creates future flux violation?”

Aligned → penalize the field.

Orthogonal → allow it — the flow is harmless.

PART 5

Implementation

Practical estimation · Algorithm 1 · Honest limits

ALGORITHM 1, ONE ITERATION

Practical Estimation

p_data and its score are unknown — approximate ∇log p with a minibatch KDE, then:

1. **Sample x_0** › **2.** **KDE score ∇log p** › **3.** **Langevin $x_0 \to x_t$** › **4.** **Compute uθ, rθ** › **5.** **Apply L_flux**

ALGORITHM 1 — ONE TRAINING ITERATION OF FLUX MATCHING

Algorithm 1 One Training Iteration of Flux Matching.

$\boldsymbol{x}_0$ denotes initial samples, $\boldsymbol{x}_0^{(i)}$ highlights the selected initial sample, $\boldsymbol{x}_t$ denotes simulated samples at time t, and $\boldsymbol{w}_{im}$ denotes the weight of simulated sample m on $x_0^{(i)}$. *(Note: Font colors match the visual nodes and edges in Figure 3).*

Input: minibatch $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^B$, bandwidth σ, learnable vector field f_θ

1: Construct $\nabla \log \widehat{p}_\sigma$ via Equation (7) and compute $\boldsymbol{u}_\theta(\boldsymbol{x}_0^{(i)}) = \boldsymbol{f}_\theta(\boldsymbol{x}_0^{(i)}) - \boldsymbol{\nabla} \log \widehat{\boldsymbol{p}}_\sigma(\boldsymbol{x}_0^{(i)})$ for all i
2: Sample shared simulation time $t \sim q$ on $[0, T]$ with $T = 4\sigma^2$
3: MCMC $\{\boldsymbol{x}_0^{(i)}\}_{i=1}^B$ to $\{\boldsymbol{x}_t^{(i)}\}_{i=1}^B$ with $\nabla \log \widehat{p}_\sigma$ using Equation (8)
4: For selected initial sample $\boldsymbol{x}_0^{(i)}$, calculate Equation (9): $\widehat{\partial x_t/\partial x_0^\top} \nabla r_\theta(x_0^{(i)}, t) := \sum_{m=1}^B \boldsymbol{w}_{im} \boldsymbol{\nabla}_{\boldsymbol{x}_t^{(m)}} \boldsymbol{r}_\theta(\boldsymbol{x}_t^{(m)})$
5: Apply Step 4 for every initial sample $i = 1, \ldots, B$
6: Form $\mathcal{L}_{\text{flux}}$ from Equation (6) and update θ

What Flux Matching Does Not Guarantee

Bare Flux Matching enforces only one thing: the same stationary distribution.

NOT AUTOMATIC

- ✗ Straight trajectories
- ✗ Fast sampling
- ✗ Good tail exploration
- ✗ Non-score solutions

STILL VALID SOLUTIONS

- The score field itself
- Ugly, curvy density-preserving fields

Underdetermined — correctness alone does not pick a useful field; the dynamics must be chosen.

PART 6

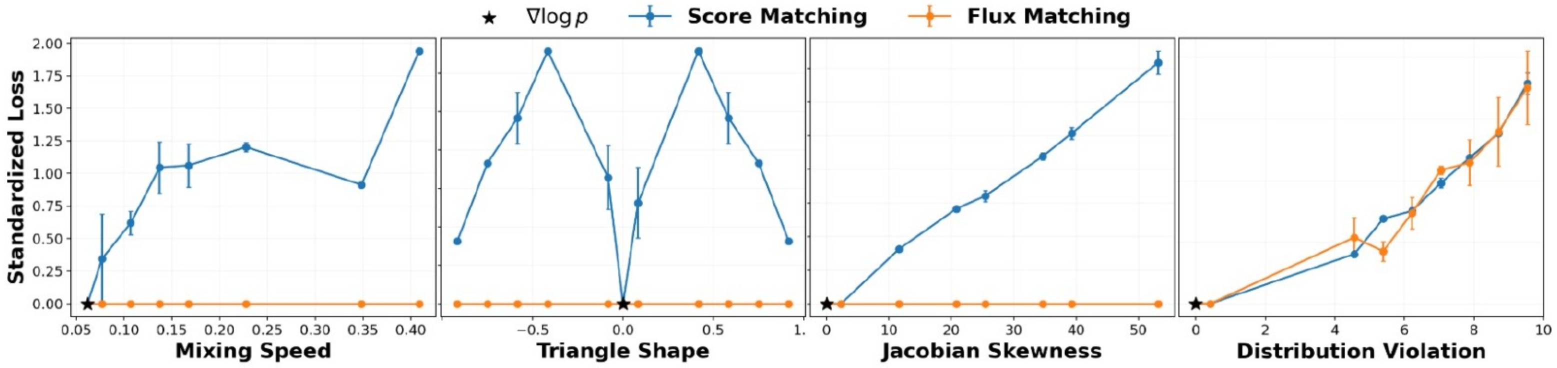
Applications & Results

Controllability · RNA velocity · Generation · Faster sampling · Directed structure

Controllable Generative Fields

On a 2D three-component Gaussian mixture, three one-parameter families of distribution-preserving fields (mixing speed, triangle shape, Jacobian skewness) plus a distribution-violating family.

FIGURE 4 — SCORE MATCHING VS. FLUX MATCHING LOSS



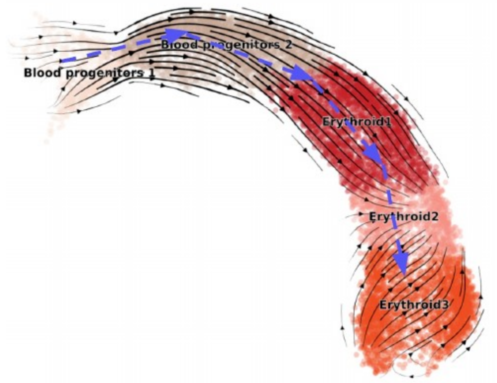
KEY RESULT

Score matching's loss rises with any perturbation; Flux Matching stays exactly zero across all distribution-preserving changes — and only rises under genuine distribution violation.

Interpretable Fields: RNA Velocity

Keep the biophysical ODE unchanged; replace its bespoke EM fit with gradient descent on L_flux across G = 2000 genes — Flux Matching supplies the training objective.

FIGURE 5 — LEARNED VELOCITY



CROSS-BOUNDARY CORRECTNESS / CONSISTENCY (↑)

Dataset	Flux Matching	scVelo [5]
Pancreas	0.202 / 0.972	0.330 / 0.821
Gastrulation	0.611 / 0.991	−0.639 / 0.877
Dentate Gyrus	0.284 / 0.981	−0.084 / 0.791
Bone Marrow	0.177 / 0.939	−0.789 / 0.857
Hindbrain	0.345 / 0.897	0.332 / 0.874

KEY RESULT

Same model class: Flux Matching improves consistency on all five datasets and CBC on four of five vs. scVelo.

Standalone Image Generation

Noise-annealed Flux Matching as a standalone objective on CIFAR-10 and CelebA 64×64, standard UNet; DSM baseline shares architecture and hyperparameters.

GENERATION QUALITY

Dataset	Model	FID ↓	IS ↑	NLL ↓
CIFAR-10	DSM	4.74	8.52	3.16
	Flux	9.06	8.54	3.26
CelebA	DSM	2.41	–	2.03
	Flux	7.07	–	2.17

TRAINING EFFICIENCY

Dataset	Model	Speed it/s ↑	Mem G ↓
CIFAR-10	DSM	11.63	2.79
	Flux	4.01	5.69
CelebA	DSM	7.20	7.79
	Flux	1.77	22.67

KEY RESULT

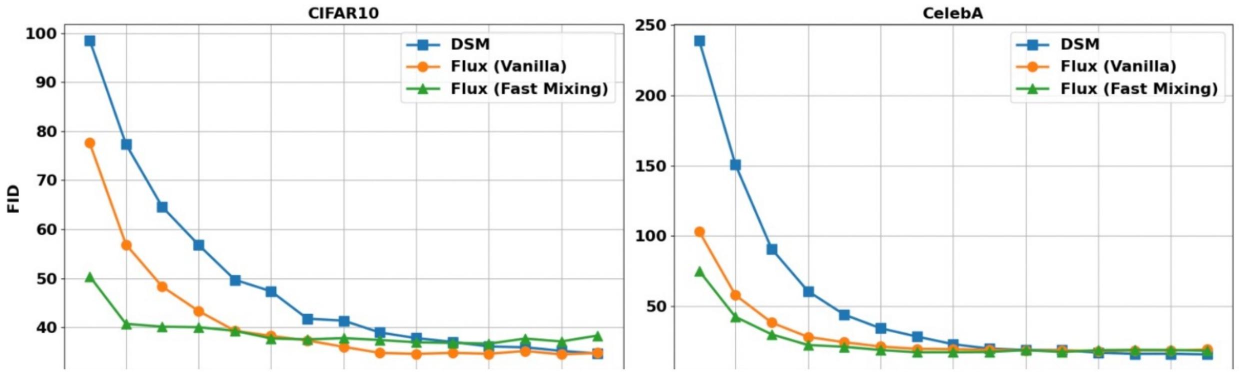
Flux Matching is competitive on IS and NLL; the FID gap reflects DSM’s heavy FID-specific engineering. As a first-pass new objective, it trains ~3–4× slower and uses more memory.

[Table 1 · §4.3] unconditional generation

Faster Sampling via Fast-Mixing Fields

Add a fast-mixing proxy loss (λ = 0.01): L = L_flux-noise + λ L_mixing, same model as §4.3; evaluate FID on 1K samples vs. number of sampling steps.

FIGURE 6 — FID VS. SAMPLING STEPS



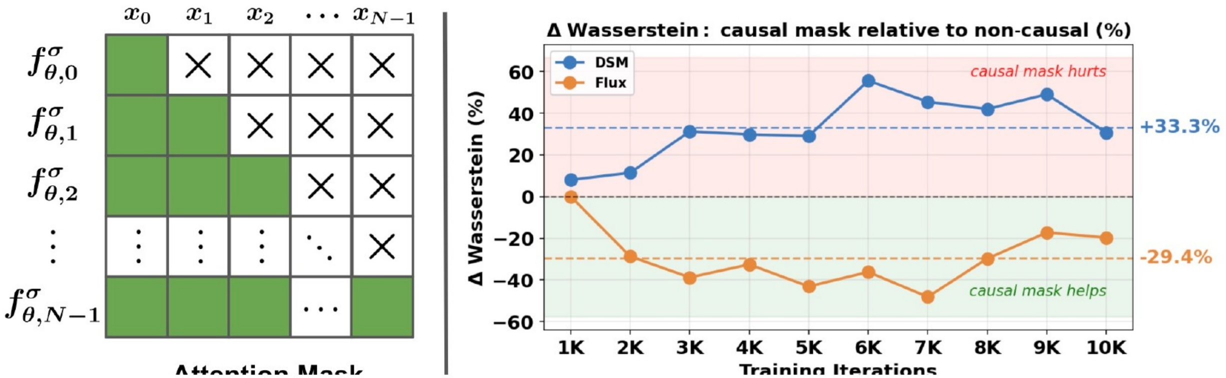
KEY RESULT

Fast-mixing Flux reaches comparable FID with substantially fewer sampling steps on CIFAR-10 — and even vanilla Flux already mixes faster than DSM.

Embedding Directed Structure

Two masses coupled by nonlinear springs, sampled as discretized trajectories over N time steps. A causal attention mask makes f_θ autoregressive; train DSM and Flux with/without it, score by Wasserstein W_2.

FIGURE 7 — CAUSAL MASK & ΔW_2 (MASK RELATIVE TO NON-CAUSAL)



KEY RESULT

The causal mask helps Flux Matching (−29.4% W_2) but hurts DSM (+33.3%): score fields force a symmetric Jacobian that conflicts with directed temporal dependence.

Conceptual Summary

1. **Score Matching**
   "Match the score field."
2. **Flow Matching**
   "Match a chosen transport path."
3. **Flux Matching**
   "Match only the probability-flux divergence."

THE CONDITION THAT DEFINES IT

$$\nabla\cdot\left(p\, f_\theta\right) \;=\; \nabla\cdot\left(p\,\nabla \log p\right)$$

Flux Matching turns the vector field itself into a design object.

SOURCE

Reference

THIS TALK IS BASED ON

Generative Modeling with Flux Matching

P. Pao-Huang, X. Qiu, S. Ermon

arXiv:2605.07319 (2026)

Fokker–Planck from the Heat Equation

↑ Back to FP slide

STEP 1 — PURE DIFFUSION GIVES THE HEAT EQUATION

$$dX_t = \sqrt{2}\, dW_t \qquad \frac{\partial p_t}{\partial t} = \Delta p_t = \frac{\partial^2 p_t}{\partial x^2}$$

Brownian motion alone spreads mass from dense regions into neighbouring sparse regions.

STEP 2 — REWRITE THE HEAT EQUATION AS PROBABILITY FLUX

$$\frac{\partial p_t}{\partial t} = -\nabla \cdot J_t \qquad \text{with} \qquad \Delta p_t = \nabla \cdot \nabla p_t$$

$$J_{\text{diff}}(x) = -\nabla p_t(x) = -\, p_t(x)\, \nabla \log p_t(x)$$

Diffusion moves probability down the density gradient — from dense to sparse.

Adding Drift, Then Combining

↑ Back to FP slide

STEP 3 — DETERMINISTIC DRIFT ADDS A CURRENT

$$J_{\mathrm{drift}}(x) = p_t(x)\, f_\theta(x)$$

$$\frac{\partial p_t}{\partial t} = -\nabla \cdot (p_t f_\theta)$$

STEP 4 — TOTAL FLUX = DRIFT + DIFFUSION

$$J_{\mathrm{total}} = p_t f_\theta \;-\; p_t \nabla \log p_t$$

THE FOKKER–PLANCK EQUATION (PAPER FORM)

$$\frac{\partial p_t(x)}{\partial t} = -\Big[\nabla \cdot (p_t f_\theta) \;-\; \nabla \cdot (p_t \nabla \log p_t)\Big]$$

Matching the score field pointwise is unnecessary — matching the divergence of the two fluxes suffices. This is the foundation of Flux Matching.